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Research on the Construction of Smart Education Platform Driven by Artificial Intelligence Technology in the New Era Higher Education Self-study Examination

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Abstract

In the context of the new era, the rapid development of Internet technology, artificial intelligence and other technologies, the field of education will enter the “era of intelligent education”. The research centers on the self-study examination of higher education, and builds an artificial intelligence-driven intelligent teaching platform, which integrates the functions of intelligent management, intelligent examination, intelligent education, intelligent learning, intelligent evaluation, intelligent service and so on. A feature extraction module is proposed based on convolutional neural network and knowledge graph, which utilizes the powerful feature extraction capability of CNN to extract the implicit features of courses and users, constructs a course resource recommendation model, and analyzes the application of artificial intelligence in the smart teaching platform. The accuracy (0.857 and 0.791) and area under the ROC curve (0.876 and 0.804) results of the model on different datasets are better than those of the comparison model, reflecting better resource recommendation performance. After using the smart education platform, more than 57% of learners have positive attitudes towards the platform, their learning experience, and their learning outcomes. The smart education platform constructed in the article can enhance learners’ interest, motivation, performance, and efficiency in learning, and help them achieve intelligence and efficiency in self-study exam preparation.

Keywords: Artificial intelligence; Knowledge graph; Convolutional neural network; Resource recommendation; Intelligent education platform.

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1 Introduction

Self-study examination for higher education refers to a part-time higher education examination system under the supervision of the Central Television University. The self-study examination is set up for people who work, expand themselves and improve their qualifications. Its distinctive feature lies in its high degree of autonomy and flexibility. Candidates can independently choose the majors and courses according to their own time, ability and interest. This autonomy allows candidates to better combine their own actual situation, develop a personalized learning plan, and give full play to their personal strengths and potential. In addition, the college, undergraduate, master's degree, doctoral courses offered by the self-study examination are diverse, and the registration conditions are relaxed, and the study time of the self-study examination is independently arranged, and there is enough time to balance work and study, so the construction of the intelligent education platform of artificial intelligence has an important role in the self-study examination [1-4].

Intelligent education platform is an important development trend in the field of education. It provides more flexible and diversified learning methods for self-study candidates by utilizing the Internet and intelligent technology. The smart education platform not only changes the traditional learning mode, but also promotes the development of education equity and personalized education. The intelligent education platform also provides rich learning resources and teaching tools, and candidates can access paper and electronic books, course videos, online teaching materials, etc. through the Internet. Compared with the traditional learning mode, the smart education platform pays more attention to the personalized needs and interests of candidates, which is conducive to improving the knowledge base of self-study candidates [5-8].

The higher education self-study examination is an important form of education in China's higher education system. As a flexible form of education, the Self-study Examination for Higher Education (SSE) provides an independent and autonomous learning pathway for students in the workplace and society who wish to further their studies. Literature [9] emphasizes the importance of self-study examinations. The demand for self-study examination information is analyzed by applying the idea of cloud computing, and a cloud-oriented information construction scheme for higher education self-study examination, i.e., self-study examination as a service, is proposed. And benefit cloud computing ideas on the architecture and application mode of cloud computing are discussed. Literature [10] examined the self-study examination of art and design majors. It is emphasized that in order to realize the cultivation goal of professional applied talents, it is very necessary to explore the education mode combining self-study examination and skill training. Teaching reform was elaborated based on changing teaching concepts and other aspects. This is not only in line with the direction of cultivating skill-oriented talents in higher education, but also has special practical significance. Literature [11] emphasized the promotion role of China's higher education self-study examination on education reform and economic discovery, based on the analysis of China's higher education talent cultivation mode, and put forward the coping strategies for its problems. Literature [12] pointed out that in recent years, the self-study examination system has made proud achievements, but with the introduction of various forms of higher education, the self-study examination has suffered a heavy blow, and its problems have been highlighted, and only by solving these problems can we realize the healthy and sustainable development of the self-study examination. Literature [13] explored the operation of educational services of self-study higher education examination. Based on questionnaires, interviews and other methods, it was concluded that the educational objectives of the self-study higher education examination are more implicit, but due to the poor applicability, low efficiency and other problems it is subjected to measures and weakened. Literature [14] states that self-study higher education examination has been well developed since its inception, but it faces many challenges in its further development. By analyzing the challenges faced by the self-study examination, strategies are proposed to cope with them.

Smart Education Platform is an education platform based on artificial intelligence technology. It mainly analyzes students' learning, behavior and performance to recommend teaching resources suitable for each student, so as to improve students' learning effect and teaching quality. Literature [15] creates a new intelligent and efficient intelligent device classroom teaching model, which combines the current situation and development of the intelligent device classroom and examines the shortcomings of traditional college language teaching. Its findings provide reference for the implementation of intelligent classroom teaching. Literature [16] introduces the AI-assisted interactive smart education framework, which aims to improve the interaction between students in higher education smart education. The framework provides students with reliable learning materials and feedback systems to analyze the learning performance of smart education. The experimental results point out that the proposed framework improves student-teacher interactions and shows high accuracy in analyzing students' academic performance skills. Literature [17] examined the design and application of a digital intelligent teaching cloud platform based on AI algorithms. And through case studies and data research, it was found that the intelligent teaching platform helps to improve teaching efficiency and students' learning experience, but there are also challenges such as data privacy. Ideas are provided for the development of intelligent teaching cloud platform. Literature [18] used interviews and case studies to examine the new ideas of AI in the elements of teaching and learning activities, which lie in the promotion of the intelligent development of education. By applying AI in physical education, personalized teaching was achieved by constructing Agent layer and data service layer. Comparative experiments yielded that the system improved students' physical education performance and sports interest. Literature [19] aims to develop a framework for educational AI learning platforms and evaluate its applicability. One of the AI educational learning platform frameworks was found to have particularly good applicability evaluation results. Literature [20] built an AI-based personalized intelligent learning service platform that improves the efficiency of users and provides open, on-demand education where knowledge can be accessed from resources at any time, which is realized to promote the development of scholars.

Based on the self-study examination of higher education, the application architecture of the intelligent education platform is proposed from the realization of functions such as management services, examination and teaching affairs, and learning evaluation to realize the information sharing of learning resources and improve the convenience of management and services. At the same time, considering the integration of artificial intelligence technology into the platform application, a course resource recommendation algorithm based on knowledge graph and convolutional neural network is proposed, using the convolution-based feature extraction model module to deeply mine the historical information of the interaction with the user, extracting feature vectors representing the attributes of the course as well as feature vectors representing the attributes of the user to complement the vectors of the original user as well as the course, and conducting more angles of the Higher-order aggregation between the vectors, so as to obtain more accurate embedded vector representations of users and courses. Different datasets and resource recommendation models are selected for comparison experiments to explore the effect of course resource recommendation on the proposed KGCNN model. Finally, a questionnaire survey on learners' feedback on the use of the smart education platform is conducted from the three dimensions of platform satisfaction, learning attitude and learning effect, so as to test the actual application effect of the smart education platform.

2 Construction of a smart education platform based on self-study exams

Self-study examination is a form of higher education that combines individual self-study with national examination, mainly recruiting socially-typed candidates. Combined with the current status quo of the informationization construction of higher education self-study examination, the

intelligent education platform construction architecture scheme is proposed from the realization of management services, examination and teaching affairs, learning and evaluation functions, etc. The application architecture of the intelligent education platform is shown in Fig. 1, and the specific functional modules of the platform are as follows.

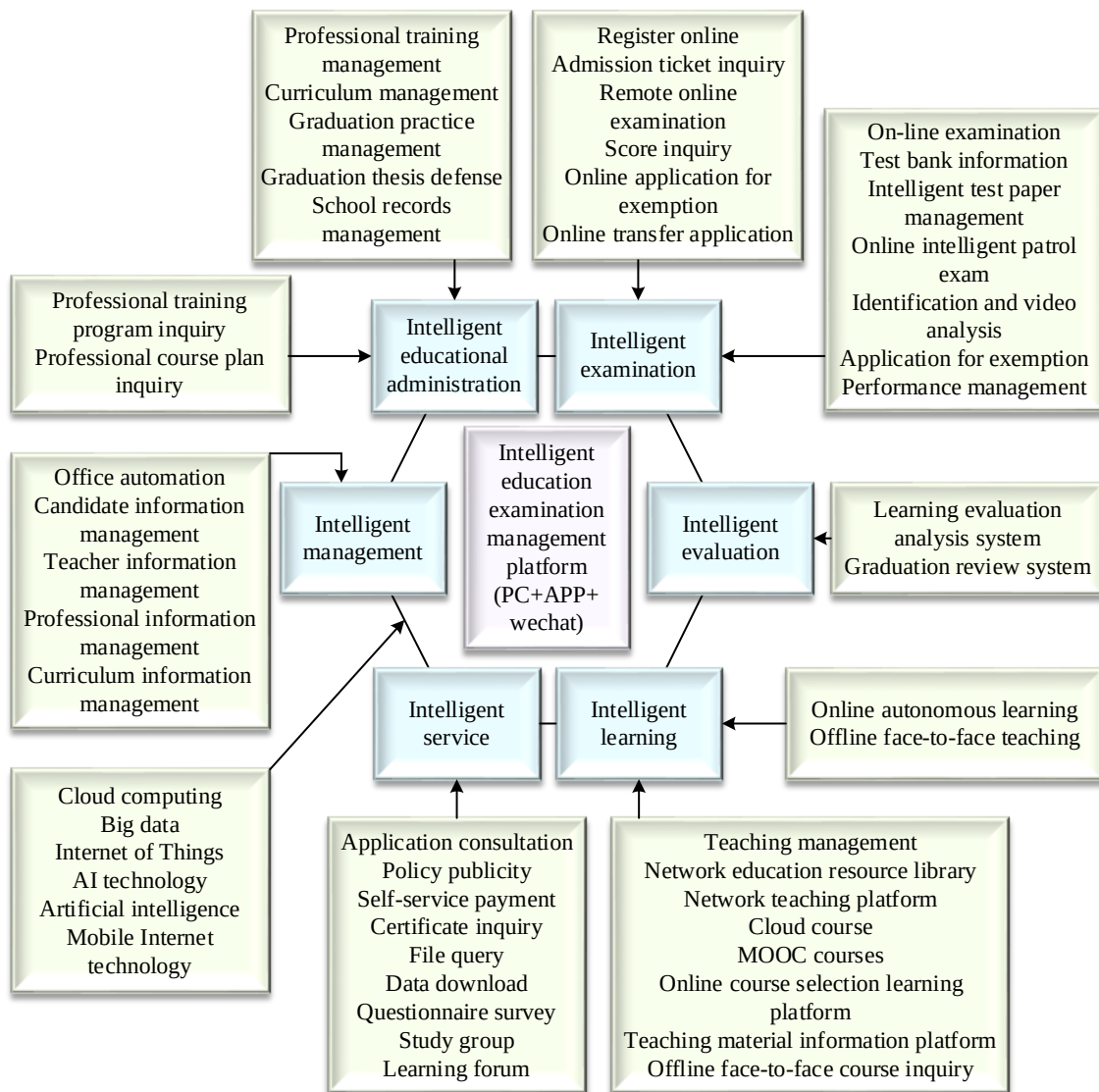


Figure 1. The application architecture of the intelligent education platform

2.1 Intelligent management

According to the work demand and workflow of the Education Examination Center, the construction system includes office automation system module, candidate information management system module, teacher information query management system module, professional information query management system module, course information query management system module, etc., to assist in the management of various types of information such as candidates, teachers, professions, courses, etc., as well as the business coordination with the colleges and universities, learning centers, study aids, etc., to achieve all kinds of office approval processes online to achieve “paperless” office. It also realizes online processing of all kinds of office approval processes, so as to achieve “paperless” office and improve the level and efficiency of the management of higher education self-study examinations.

2.2 Intelligent Examination

The platform is equipped with functions such as online application, pass inquiry, remote online examination, result inquiry, online exemption application, online transfer application to meet the needs of candidates, and at the same time, it builds an integrated intelligent examination management system to adapt to the actual needs of the examination management and make the management of higher education self-study examination more intelligent and efficient.

2.3 Intelligent Academic Affairs

Self-study exams are mainly self-study, the part is mainly to integrate resources for candidates to provide professional, curriculum and other types of teaching information query services, so this part of the function to achieve the focus is to facilitate the education and examination administrators to provide candidates with a more professional through the construction of professional training management, curriculum management, graduation practice management, graduation dissertation defense management, management of student records as one of the teaching service system, It facilitates educational examination administrators to provide candidates with more professional, accurate and efficient academic information services.

2.4 Smart Learning

To provide self-study students with intelligent learning services mainly in the form of online learning services, supplemented by offline face-to-face teaching, through the construction of network teaching management, network education resource library, network selection of online learning platforms, network teaching service platforms, cloud courses, MOOC courses, teaching materials and teaching information service platform, offline face-to-face course information service platforms and other multi-functional platforms for the integration of the intelligent learning and education service system. The system integrates and optimizes the educational and teaching resources of higher education institutions, student aid organizations and social education institutions, expanding the learning channels of self-study students in terms of time and space, and making learning more independent and efficient.

2.5 Intelligent evaluation

Provide managers and candidates with an evaluation system of candidates' learning situation, through linkage with several other platforms, real-time collection and acquisition of data on candidates' learning situation, practical internships, graduation theses, teachers' evaluations, etc., combined with the professional training program, intelligently analyze the comprehensive situation of candidates' academics, timely analyze and discover potential problems of candidates' academics, accurately provide personalized learning programs and learning suggestions, and timely push the proposed graduation candidates' reports to the managers. It also sends the report of candidates to be graduated to the administrators in time and handles them in a timely manner to ensure that the candidates graduate on time.

2.6 Intelligent Services

As the basic module of the intelligent system, it mainly provides candidates with convenient services such as application consultation, policy propaganda, self-service payment, pass query, file query, data download, questionnaire survey, study group, study forum, and so on, and conveniently

and effectively solves the practical needs encountered by self-testing students in the process of examination and study by concentrating all kinds of services in the platform of online office hall.

3 Recommendations for artificial intelligence-driven platform course resources

The Higher Education Self-study Examination's intelligent education platform uses artificial intelligence technology to provide interactive and efficient education services. Through Internet data statistics and intelligent analysis, the platform can grasp learners' cognitive development trends, knowledge application, and comprehension ability, and provide diagnostic evaluation. At the same time, the platform can analyze the characteristics of learners and push relevant learning content and programs to them, truly achieving tailor-made education from a technical perspective. This chapter proposes a platform course recommendation algorithm (KGCNN) based on knowledge graph and convolutional neural network, which better realizes course resource recommendations by mining the deep information hidden in user interaction records.

3.1 KGCNN algorithm framework

The general framework of the KGCNN algorithm is shown in Figure 2. The algorithm uses the feature extraction module to update the embedding vectors of the user and the course, then performs message aggregation and link propagation along the links in the knowledge graph, and then uses the embedding vectors of multiple layers to fuse to obtain higher-order information, and finally splices the embedding vectors of the user and the course obtained from each layer to obtain the final embedding vectors of the user and the course, and finally does the inner product of the embedding vectors of the user and the course to get the recommendation result for the user to treat the recommended course.

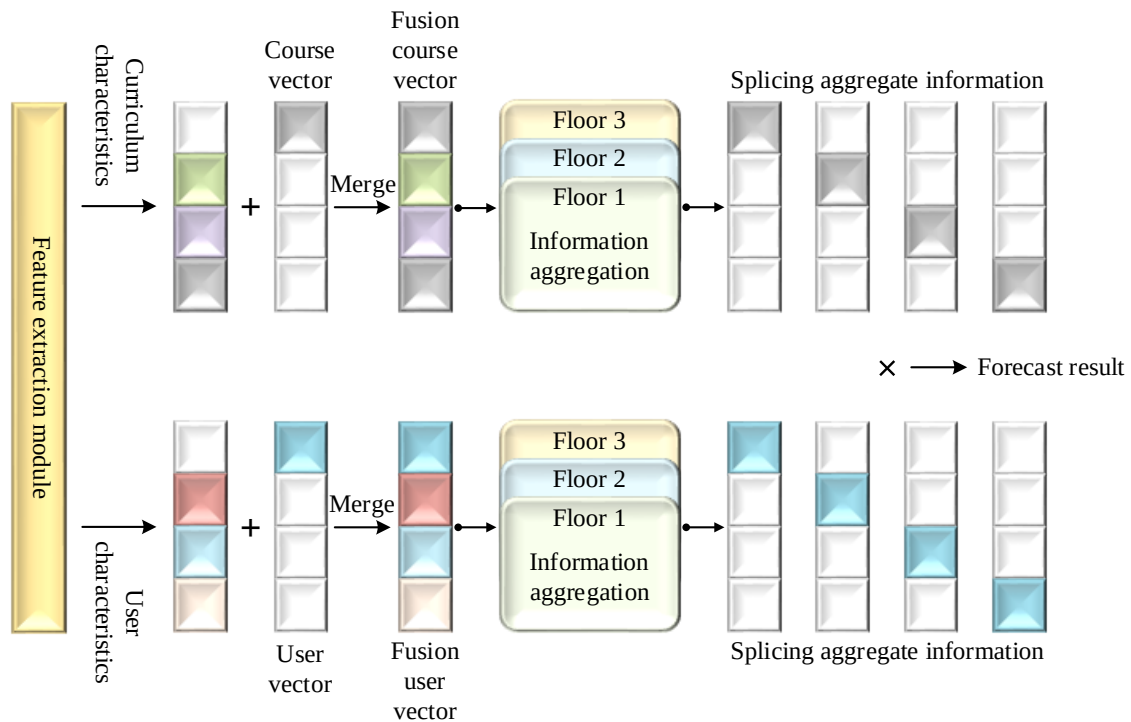


Figure 2. The overall framework of the KGCNN algorithm

3.2 Convolutional Neural Networks

Convolutional Neural Network (CNN) is one of the most representative neural networks in the field of deep learning. Convolutional Neural Network (CNN) uses a layered approach to extract features, where each layer represents the dimensions of the tensor in a three-dimensional matrix (H, W, C), which is called the feature map. Where H denotes the size or dimension of the tensor along the vertical direction, W denotes the size or dimension of the tensor along the horizontal direction and C denotes the number of feature channels contained in the tensor. The feature map is a superposition of C tensors of size $H \times W$, each representing the spatial distribution of one image feature. The input is the feature map and the output vector to achieve classification. Each component of the output vector corresponds to a feature type, indicating the probability that the recognized object belongs to that feature.

Convolutional neural networks contain three main layer structures: convolution, pooling, and fully connected. These layers are connected to each other through activation functions to construct complex convolutional neural networks. The neuron layer in a convolutional neural network consists of neurons in three dimensions, i.e., the spatial dimensions of the input: height, width, and depth.

1) Convolutional Layer

The convolutional layer is one of the core elements in a convolutional neural network and is used to extract features from the input data. Strictly speaking, the convolutional layer is a misnomer, as the operations it expresses are actually inter-correlation operations, relying on dot-multiplication of matrices for summation rather than convolutional operations.

The parameters of the convolutional layer include, given an input image I and a convolutional kernel K , a boundary padding size, a step size, and assuming that the input image has a size of $W_{in} \times H_{in} \times D_{in}$ and the size of the convolutional kernel is $k_w \times k_h \times D_{in} \times D_{out}$ (where K is the size of the convolutional kernel, D_{in} is the depth of the input image, and D_{out} is the depth of the output feature map), the output feature map o has a size of $W_{out} \times H_{out} \times D_{out}$:

$$O_{i,j,k} = \sum_{m=0}^{k_w-1} \sum_{n=0}^{k_h-1} \sum_{l=0}^{D_{in}-1} I_{i+m,j+n,l} \times K_{m,n,l,k} \quad (1)$$

Where $O_{i,j,k}$ is the value of the i, j rd position of the k nd channel of the output feature map, $I_{i+m,j+n,l}$ is the value of the $i+m, j+n$ th position of the l th channel of the input image and $K_{m,n,t,k}$ is the weight of the m, n th position of the k th channel of the convolution kernel.

Filling is a way to deal with the problem of losing edge pixels when applying multilayer convolution, mainly by performing a complementary zero operation on the image edges. When this paper is supplemented with p_h rows and p_w columns, the output shape will be:

$$(W_{in} - k_w + p_w - 1) \times (H_{in} - k_h + p_h - 1) \quad (2)$$

Further, in calculating the inter-correlation, the convolution window starts from the upper left corner and slides down and to the right, by default sliding one element at a time. By setting the step

size to slide multiple elements at a time, efficient computation and reduced sampling can be achieved. For vertical step s_h and horizontal step s_w , the output shape is:

$$[(W_{in} - k_w + p_w + s_h) / s_h] \times [(W_{in} - k_w + p_w + s_w) / s_w] \quad (3)$$

2) Pooling layer

The pooling layer is a kind of downsampling layer in the convolutional neural network used for downsampling and feature compression of the input data. The pooling layer extracts the main features in the feature map by spatially downsampling the feature map output from the convolutional layer, reduces the number of parameters in the model to reduce the computational cost and memory consumption, and helps the subsequent layers of the network to learn and represent the data features more effectively.

Common types of pooling include maximum pooling and average pooling. Maximum pooling operation is to select the maximum value as the output within a given pooling window. While average pooling operation is to calculate the average value within the pooling window and use it as the output.

3) Fully connected layer

Each neuron in the fully connected layer is connected to all the neurons in the previous layer, and each connection has a weight parameter, so the number of parameters in the fully connected layer is large. The fully connected layer is located in the last layers of the model for performing classification or regression tasks, and its role is to flatten the features extracted from the previous convolutional layer or other feature extraction layers and weight and sum all the feature connections, and then pass them to the Softmax classifier.

4) Softmax layer

In deep learning, the Softmax layer is characterized by mapping each element in the input vector to a probability value such that the sum of all probability values is 1. It is usually used as the output layer of a neural network model for multiple classification tasks. It is able to output the probability distribution for each category thus helping to determine the final classification result.

The Softmax layer formula is as follows:

$$softmax = \frac{e^{n_j}}{\sum_{j=1}^K e^{n_j}} \quad (4)$$

where n_j denotes the j nd element of the original output vector and K is the total number of categories.

3.3 Convolution-based feature extraction module

In the course recommendation domain courses can be composed of segmented attributes such as videos owned by the course, knowledge points related to the course, the domain to which the course belongs, the publisher of the course, and the main instructor of the course. In order to better mine

the deeper user and course feature information hidden under the record of user-course interaction, this paper proposes a feature extraction module based on convolutional neural network, which is designed to extract course and user features.

In this paper, the dimension of the input vector is 64, and the first layer of the module is a convolutional activation layer with 1 input channel and 3 output channels, and each convolutional layer is followed by an activation layer, and in this paper, the ReLU function is used as the activation function. A maximum pooling layer is used after the first convolutional activation layer and the sliding window size is set to 2. Immediately after this, the module uses a second convolutional layer with a number of input channels of 3 and a number of output channels of 3. A pyramid pooling layer is added before the fully-connected layer as the total number of videos, knowledge points, and domains owned by different courses is not exactly the same. This extracts the features of the embedding vectors from different perspectives while unifying the output sizes, and finally the features output from the pyramid pooling layer are mapped using the fully connected layer to obtain the final embedding vector representations of the courses and users.

3.4 KGCNN aggregation layer

Using the interaction graph between users and courses, we can obtain higher-order information between users and other courses, and use this higher-order information to model the embedding vectors of users and courses more effectively. Considering the following cases, user u_1 has studied course i_0 and user u_2 has studied i_0 and i_1 , although u_1 has not studied course i_1 but u_2 who has similar interest with user u_1 is interested in course i_1 , some features of course i_1 can be considered as contributing to the modeling of user u_1 preference. Therefore, inspired by the KGAT algorithm, the information aggregation layer used in this paper uses a fusion of multiple aggregation approaches. Define the set of embedding vectors for users as e_u , the set of embedding vectors for courses as e_i , and p_e as the splicing of user vectors and course vectors with the following formula:

$$p_e = e_u \parallel e_i \quad (5)$$

where $\parallel$ is the splice-by-row operation. Definition p_e^n is the layer n aggregation vector of the information aggregation layer, where only neighboring nodes are aggregated at each aggregation, where the information aggregation formula is as follows:

$$p_e^n = f_{relu}(f_u(p_e^{n-1} + p_s^{n-1}) + f_u(p_e^{n-1} \parallel p_s^{n-1})) \quad (6)$$

where $\parallel$ is the inner product operation, $f_{relu}(\cdot)$ is a nonlinear function, and p_e^{n-1} and p_s^{n-1} are related layer $n-1$ aggregation vectors, where $f_u(\cdot)$ is formulated as follows:

$$f_u(x) = wx + b \quad (7)$$

where w is the weight parameter and b is the corresponding bias. The corresponding p_s^n equation is as follows:

$$p_s^n = A * p_e^n \quad (8)$$

Where A is the record of the user's interaction with the course, the fusion of information between different entities is achieved by left-multiplying the interaction matrix A by the entity vector p_e^n , while p_e^0 is the corresponding initial p_e vector, which is formulated as follows:

$$p_e^0 = ((e_u + \alpha e_u^{cnn}) \square (e_i + \beta e_i^{cnn})) \quad (9)$$

Where e_u^{cnn} and e_i^{cnn} are the feature vector representations extracted by the CNN-based feature extraction module for users and courses, respectively, and α and β are the corresponding weight parameters, both of which are 0.4 in this paper's algorithm α and β .

3.5 KGCNN Loss Function

BPR loss is a loss function used in recommender systems, which is based on the Bayesian Personalized Ranking (BPR) model. The BPR loss is used in KGCNN algorithm as the optimization objective of the algorithm, which is formulated as follows:

$$Loss_{BPR} = \sum_{(u,i,j) \in O} -\ln \sigma(\hat{y}_{ui} - \hat{y}_{uj}) + \lambda \|\theta\|^2 \quad (10)$$

where (u, i, j) is two sets of training data, i is the course that user u has actually interacted with, j is the course that user has not interacted with, $\hat{y}_{ui}$ is the corresponding user u 's prediction score for course i , and $\|\theta\|^2$ is the regularization term, which generally prevents the model from overfitting.

3.6 Experimentation and Analysis

3.6.1 Data sets

Deep learning algorithms require more data than traditional machine learning algorithms. In order to meet this demand, this chapter adopts the MOOCCube and Criteo datasets, which are derived from the open datasets of the Schoolhouse Online website.

MOOCCube is a free data warehouse that provides access to information about 706 courses and hundreds of thousands of course selection records from 190,000 users in online education. According to the description on the official website of MOOCCube dataset, the dataset can be used in several MOOC-related research areas such as course recommendation, student behavior prediction, and course concept extraction. In this chapter, the CourseRecommend dataset, obtained after appropriate processing of the MOOCCube dataset, is used as the data support for course recommendations. The ratio of the divided training set, test set, and validation set is 8:1:1.

Criteo, a digital company specializing in online performance marketing, makes publicly available the Criteo dataset, which is widely used to evaluate the performance of recommender systems. In this experiment, 1 million pieces of data were randomly selected from the huge Criteo dataset and rigorously divided into 10% for the test set, 10% for the validation set, and 80% for the training set. Such a division method can effectively simulate actual application scenarios, and model training and performance evaluation can be carried out on this basis.

3.6.2 Contrasting models

In order to evaluate the performance of the KGCNN model proposed in this paper, it is validated using comparative experiments using the following comparative models: the Wide&Deep model, the DeepFM model, the DCN model, and the XDeepFM model.

3.6.3 Analysis of experimental results

For the course resource recommendation experiments on the smart education platform, this paper selects the area under the ROC curve and the accuracy rate to evaluate the experimental results. Fig. 3 and Fig. 4 show the accuracy and loss function changes of the KGCNN model after 200 rounds of training in the CourseRecommend dataset, respectively, and the number of horizontal coordinate rounds in the figure represents the number of rounds of training. As the number of rounds increases, the accuracy of the training and validation sets is increasing, the loss function is decreasing, and the variation decreases after about 50 rounds of training, and the overall phenomenon of convergence is shown.

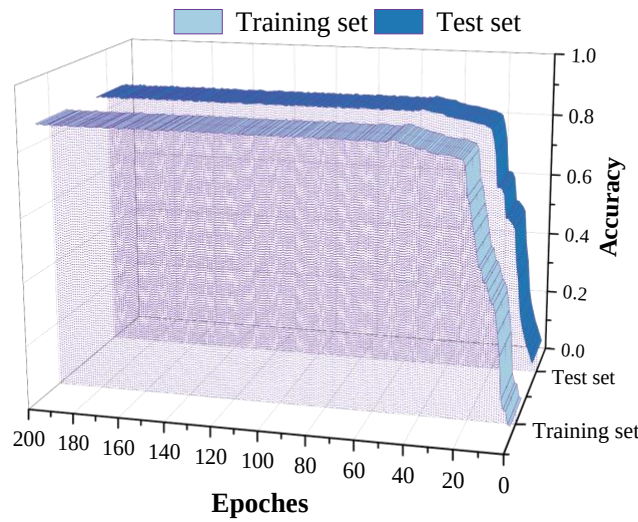


Figure 3. The accuracy of the KGCNN model changes

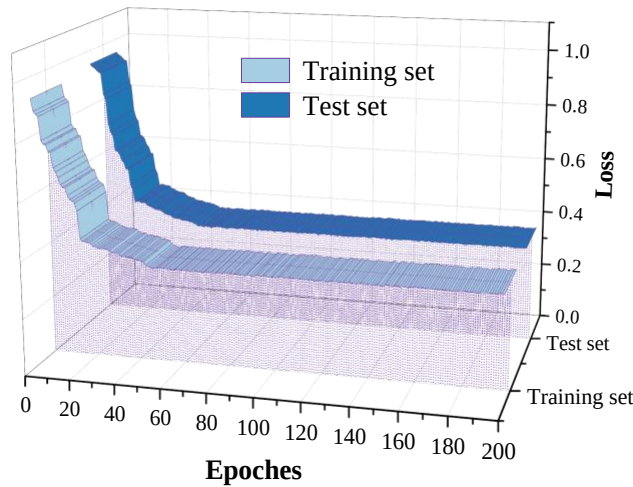


Figure 4. The loss of the KGCNN model changes

Figure 5 shows the accuracy results for each model on the CourseRecommend dataset and Criteo dataset, and Figure 6 shows the area under the ROC curve results for each model on the CourseRecommend dataset and Criteo dataset. The Wide&Deep model, which is the base model, is generally lower in terms of performance than the other models. Whereas, the XDeepFM model and DCN model, which use higher-order display feature interactions, outperform the DeepFM model and Wide&Deep model, which do not use higher-order display feature interactions. The accuracy of the KGCNN model proposed in this chapter on CourseRecommend dataset and Criteo dataset is 0.857 and 0.791, and the area under the ROC curve is 0.876 and 0.804, respectively, which have the optimal accuracy and area under the ROC curve in the experimental results. The KGCNN model in this paper has better course resource recommendation performance in the smart education platform based on self-study exam, and it can recommend learning resources for higher education self-study exam students in line with their preferences.

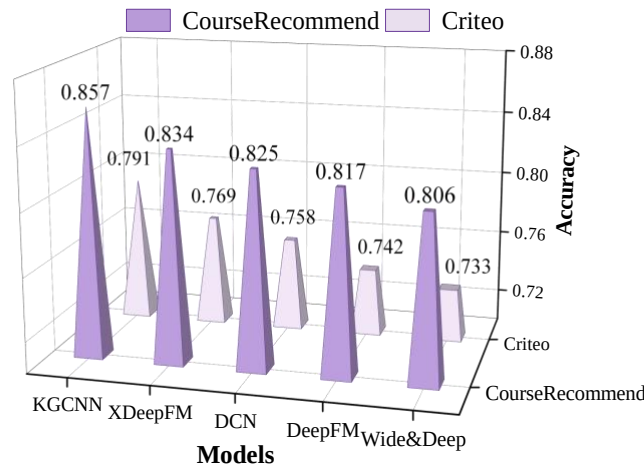


Figure 5. Comparison of the accuracy of different models

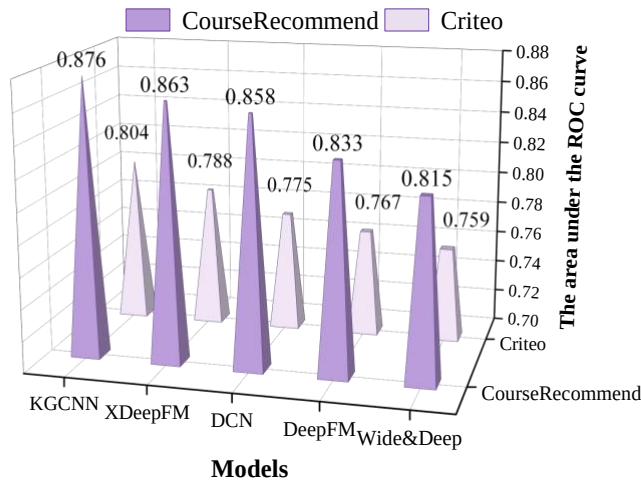


Figure 6. Comparison of the ROC curve of different models

4 Evaluation and analysis of the Intelligent Education Platform

The main purpose of this study is to build a smart education platform for self-study exams in higher education. After the design and realization of the functional requirements for self-study exams are completed, the smart education platform has to test the application effect of the system and modify the problems of the platform according to the test results to further improve the system. In order to

verify whether the intelligent education platform improves the learning efficiency of learners and recommends appropriate course learning resources according to the needs of learners to a certain extent, the application effect of the platform is verified through the questionnaire survey method.

4.1 Questionnaires

The target of this questionnaire was selected from freshman to sophomore students of University of M. Feedback was gathered from learners after using the platform. The questionnaire on the use effect of the smart education platform was designed to evaluate the system evaluation indexes from the three dimensions of satisfaction with the platform, learning attitude, and learning results. The total number of questionnaires distributed was 194 and only 182 were effectively collected.

4.2 Analysis of application effects

4.2.1 Analysis of platform satisfaction

In the questionnaire on the use effect of the smart education platform, questions 1, 2 and 3 of the survey questions analyze the satisfaction survey of the self-study exam learners with the platform, mainly from the three aspects of whether the learners support the behavioral data to be accessed and used, whether the platform's function is designed to be comprehensive, and whether the recommended resources are in line with the personalized needs. The results of the satisfaction survey of the platform are shown in Figure 7, with A~E corresponding to very compliant, relatively compliant, generally compliant, not very compliant, and very non-compliant (same as below).

23.1% of learners strongly support the use of learner behavioral data, 38.9% support the use of behavioral data, and the majority of learners support the use of behavioral data, to varying degrees, to analyze learners' personalized characteristics and recommend learning resources for them. 25.1% of learners believe that the platform functionality is very comprehensive, and 32.8% believe that the system's resources are more comprehensive, and, in general. Overall, most learners think that the platform functions in the system are comprehensively designed, but a few learners think that more can be added. Meanwhile, 30.9% and 35.4% of the learners think that the course resources recommendation meets the needs to different degrees, indicating that most learners think that the course resources recommendation function of the platform can basically meet the personalized needs of the learners, and has a good recommendation effect. In general, most learners acknowledge the satisfaction of the smart education platform to varying degrees, and the learner satisfaction survey analysis indicates that the smart education platform has certain effectiveness.

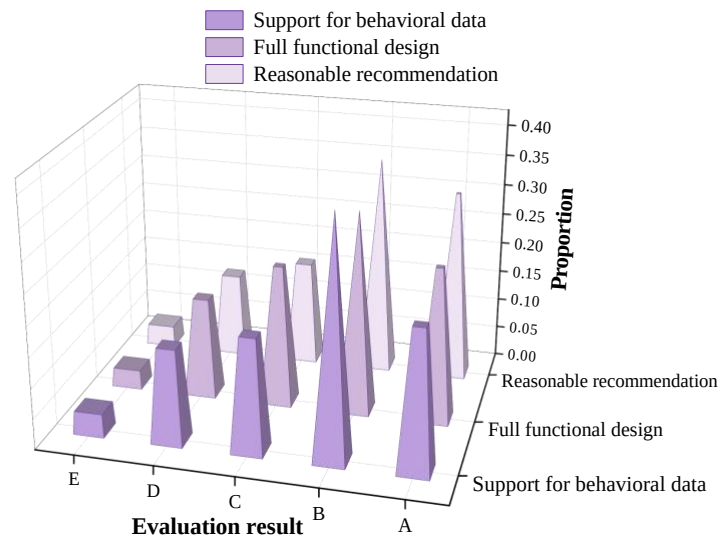


Figure 7. Results of platform satisfaction survey

4.2.2 Analysis of learning attitudes

The 4th, 5th, and 6th questions in the questionnaire survey analyze the learners' learning attitudes after having used the Smart Education Platform, mainly from the three aspects of whether the system is a powerful tool to assist self-study exams, whether the system can improve learning interest, and whether the system improves learning motivation.

The results of the survey on learning attitudes after using the platform are shown in Figure 8. 26.2% of the learners and 35.4% of the learners thought that the smart education platform was a better tool to assist self-study exams to different degrees, and overall the platform could help learners prepare for self-study exams to a certain extent. 32.4% of the learners strongly agreed that the platform could increase learners' interest in learning, and 38.2% of the learners think that the platform can improve learning interest, and there are a few learners who think that the platform can't help learners to improve their learning interest. 22.6% and 39.3% of learners agree to varying degrees that the platform can improve learning motivation, and in general, the intelligent education platform has a good and positive effect on self-study exams.

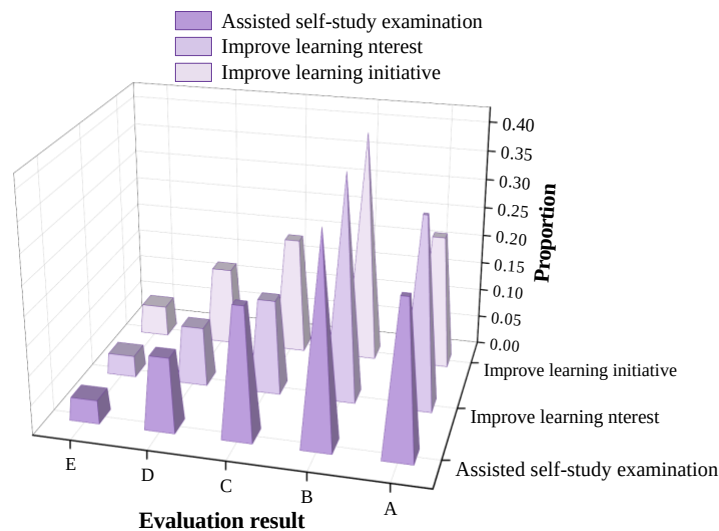


Figure 8. The results of the study attitude survey after the platform use

4.2.3 Analysis of learning outcomes

Questions 7 and 8 of the questionnaire are entitled Analyzing Learners' Learning Effectiveness Survey of the Smart Education Platform, which is mainly analyzed in terms of whether the platform can help learners improve their grades and whether the platform can improve their learning efficiency.

The results of the survey on the learning effect of the platform after its use are shown in Figure 9. The survey results show that 31.1% of the learners and 38.3% of the learners think that the platform can help them improve their academic performance to different degrees. The platform's effectiveness in improving learning efficiency was recognized by 64.0% of learners. A small group of learners believe that the platform has no impact on the learning of self-study exams. Overall, learners recognize that the intelligent education platform has a certain role in helping learners prepare for self-study exams.

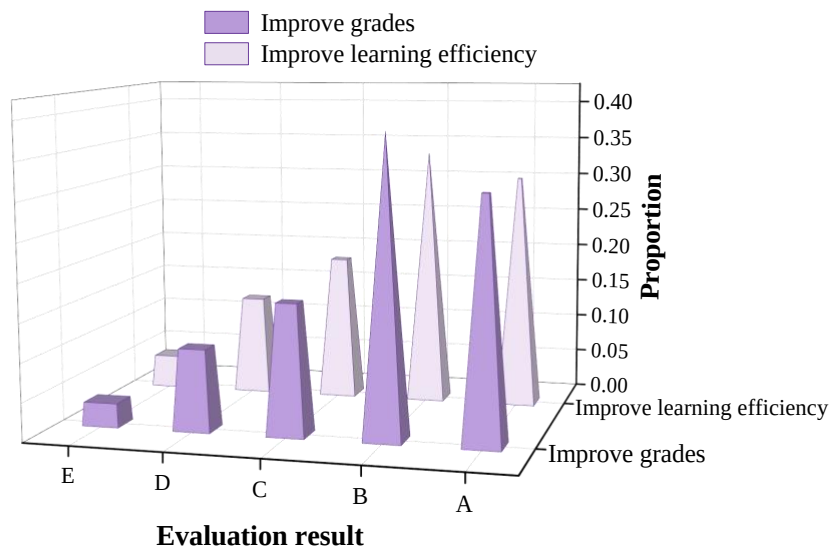


Figure 9. The results of the study effect of the platform

5 Conclusion

With the progress of science and technology, utilizing intelligent teaching platforms for learning is an inevitable trend in the process of students' self-study examination. Based on the self-study examination of higher education, the study constructs a smart education platform that integrates the functions of management, learning, evaluation, and service. Combined with knowledge graph and convolutional neural network, the course resource recommendation model available on the smart education platform is proposed. Finally, the smart education platform is evaluated through a questionnaire survey.

The study constructs a course resource recommendation model to analyze the use of artificial intelligence in the smart education platform. Through experiments, it is found that the KGCNN model proposed in this paper has good recommendation accuracy and convergence effect, and the accuracy and area under the ROC curve are higher than the comparison model on CourseRecommend dataset and Criteo dataset, in which the accuracy is 0.857 and 0.791, and the area under the ROC curve is 0.876 and 0.804, respectively.

The smart education platform passed validation of effectiveness, and over 57% of learners had a positive attitude towards the platform in the three dimensions of satisfaction, learning attitude, and learning effect. After applying the wisdom education platform for self-study exam learning, learners' interest in learning, learning motivation, learning performance and learning efficiency are all improved, and they think that the platform's course resource recommendation is more reasonable and is a better self-study exam support tool.

The establishment of an open and flexible learning platform for self-study exams with strong applicability is not only to meet the learning demands of the candidates and adapt to the needs of education informatization, but also the inherent needs of the reform and development of the self-study exam system itself. The application of the intelligent education platform to the self-study examination has changed the traditional self-study examination learning mode, and the diversified learning resources can mobilize students' enthusiasm and make the self-study examination learning more efficient.

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